Lecture 18

Introduction to Second Order Sliding Mode

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Sliding mode control and its application

Outline

1. Revision of the Sliding Mode Control
2. Why Second Order Sliding Mode
3. Second Order Sliding Mode Control
4. Proof of Twisting Controller

Sliding mode control: Revised

Consider the Linear System

$$\dot{x} = Ax + B(u + \rho(t, x))$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}$.

- The sliding surface is designed as $\sigma(x) = Cx = 0$.
- The sliding mode control is designed to reach the sliding surface in finite time.

Assumptions

- The pair (A, B), is controllable.
- The scalar $CB \neq 0$.
- The disturbance/uncertainties are matched and bounded, i.e.,$|\rho(t, x)| \leq D_M$.
- The zero dynamics (reduced order dynamics) of the system with the output $\sigma(x)$ is stable.

Controller design

Since the function $\sigma(x)$ is to be made zero, it helps to look at its derivative.

$$\begin{aligned}\dot{\sigma} &= C\dot{x} \\ &= CAx + CB(u + \rho(t, x))\end{aligned} \tag{1}$$

Now, let us design the control as,

$$u = (CB)^{-1}\left(-CAx + \phi\left(\sigma\right)\right)$$

Substituting this control in (1) gives,

$$\dot{\sigma} = \phi(\sigma) + CB\rho(t, x) \tag{2}$$

- The function $\phi(\sigma)$ must be selected such that, the equation (2) provides finite time convergence of σ to zero, in presence of $\rho(t, x)$.
- Once the sliding surface is reached, i.e., $\sigma(x) \equiv 0$, the system trajectory is governed by the zero dynamics, also called reduced-order dynamics.

Why second order sliding mode

Examples of $\phi(\sigma)$

1. $\phi(\sigma) = -K\ sgn(\sigma)$
2. $\phi(\sigma) = -\lambda\sigma - K\ sgn(\sigma)$
3. $\phi(\sigma) = -K|\sigma|^{\alpha}\ sgn(\sigma)$

However, the third choice can not reject the disturbances if $\rho(t,x) \neq 0$ at $x = 0$.

Chattering Problem

The control is discontinuous, resulting in high-frequency switching by the actuator.

Relative Degree Problem

$$\dot{\sigma} = CAx + CB(u + \rho(t,x))$$

- If $CB = 0$, then control does not appear in the expression for $\dot{\sigma}$.
- Usually, C is designed surface matrix so $CB = 0$ can be avoided.

However, σ may be some output of the system, and C is determined by system dynamics. The objective is to make the output zero robustly, then with $CB = 0$, the sliding mode control design is not possible.

Chattering removal

Adding an integrator in the control design:

Consider the second derivative of the switching function $\sigma(x)$

$$\begin{aligned}\ddot{\sigma} &= CA\dot{x} + CB(\dot{u} + \dot{\rho}) \\ &= CA^2x + CABu + CB\dot{u} + CB\dot{\rho} \end{aligned} \tag{3}$$

Let us design the control derivative as

$$\dot{u} = (CB)^{-1}(-CA^2x - CABu + \phi(\sigma, \dot{\sigma}))$$

Substituting this control derivative in (3) gives

$$\ddot{\sigma} = \phi(\sigma, \dot{\sigma}) + CB\dot{\rho}(t, x) \tag{4}$$

Question?

Does there exist a $\phi(\sigma, \dot{\sigma})$ which stabilizes (4) in finite time, if $|CB\dot{\rho}| \leq D'_M$?

Chattering removal

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Answer: Twisting Controller

$$\phi(\sigma, \dot{\sigma}) = -k_1\ sgn(\sigma) - k_2\ sgn(\dot{\sigma}),\ k_1 > k_2 + D'_M > 0$$

Relative degree

Example of Relative Degree Two Problem

- Consider some system equations derived from first principles

$$\ddot{q} = a_1 q + a_2 \dot{q} + F + \rho(t, q, \dot{q})$$

where F is external force and ρ is the disturbances/uncertainties in parameters.

- Suppose, the objective is to make the position q track some desired constant position q_d.

- Define the error as the switching function for the sliding mode control

$$\begin{aligned} e &= q - q_d, \\ \dot{e} &= \dot{q} \\ \ddot{e} &= a_1 q + a_2 \dot{q} + F + \rho(t, q, \dot{q}) \end{aligned}$$

Relative degree

- Note that the control input does not appear in the expression of $\dot{e}$. Thus, the $e = 0$ cannot be used as a sliding surface.
- Next, design the control force as

$$F = -a_1 q - a_2 \dot{q} + \phi(e, \dot{e})$$

- Substitution gives the following error dynamics,

$$\ddot{e} = \phi(e, \dot{e}) + \rho$$

- The twisting controller can stabilize in the presence of bounded uncertainty ρ.

Some concepts to understand proof of "Twisting Controller"

Homogeneous Functions

A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be homogeneous of degree m if

$$f(\lambda x) = \lambda^m f(x), x \in \mathbb{R}^n$$

for any $\lambda > 0$. Here, $\lambda x = (\lambda x_1, \cdots, \lambda x_n)$, i.e, each component of the vector x is multiplied by the same constant.

Weighted Homogeneity

A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be weighted homogeneous of degree m if

$$f(\Lambda x) = \lambda^m f(x), x \in \mathbb{R}^n$$

for any $\lambda > 0$. Here, $\Lambda x = (\lambda^{p_1} x_1, \cdots, \lambda^{p_n} x_n)$, i.e, each component of the vector x is given different weights. Specifically, exponents of the same constant.

Dilation

Formally, Λ is called a *dilation*. It is a mapping from $\mathbb{R}^n$ to itself.

$$\Lambda : (x_1, \cdots, x_n) \mapsto (\lambda^{p_1} x_1, \cdots, \lambda^{p_n} x_n)$$

Examples of homogeneous functions

Determine if the following functions are homogeneous (weighted homogeneous). Also, determine the degree of homogeneity if a function is homogeneous.

$$f(x) = \frac{2x_1^3 + x_1x_2^2 - 3x_2^3}{5x_1x_2}$$

$$f(x) = \log \frac{3x_1^2x_2}{x_1^3 - 4x_1x_2^2}$$

$$f(x) = \frac{x_1x_2^3}{x_1 + x_2^2}$$

$$f(x) = \frac{x_1x_2^2}{x_1^3 + x_2^2 - x_1^2x_2^{\frac{2}{3}}}$$

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$$f(x) = \frac{2x_1^3 + x_1x_2^2 - 3x_2^3}{5x_1x_2}$$

Homogeneous, Degree=1

$$f(x) = \log \frac{3x_1^2x_2}{x_1^3 - 4x_1x_2^2}$$

Homogeneous, Degree=0

$$f(x) = \frac{x_1x_2^3}{x_1 + x_2^2}$$

Weighted Homogeneous, Degree=3
Dilation: $(x_1, x_2) \rightarrow (\lambda^2 x_1, \lambda x_2)$

$$f(x) = \frac{x_1x_2^2}{x_1^3 + x_2^2 - x_1^2x_2^{\frac{2}{3}}}$$

Weighted Homogeneous, Degree=2
Dilation: $(x_1, x_2) \rightarrow (\lambda^2 x_1, \lambda^3 x_2)$

$$f(x) = \frac{x_1x_2^2}{x_1^3 + x_2^2 - x_1^2x_2^{\frac{1}{3}}}$$

Not Homogeneous

Homogeneous vector fields

A Vector Field

Loosely speaking, the vector field is a mapping between two vector spaces, i.e., $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, then f is a vector field. The vector field can be understood as a vector of functions

$$f(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_n(x) \end{pmatrix}$$

where, each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$

Homogeneous Vector Field

A vector field f is homogeneous of degree m, w.r.t, a dilation $\Delta_\lambda(x_1, \cdots, x_n) \rightarrow (\lambda^{r_1} x_1, \cdots, \lambda^{r_n} x_n)$ if and only if for each i, the i$^{\text{th}}$ component f_i is homogeneous with degree $m + r_i$.

Homogeneous vector field

Example: Consider the following vector field to determine its degree of homogeneity and dilation. Given that a_1, a_2, $\beta \in \mathbb{R}$

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -a_1 x_1^{\frac{\beta}{2-\beta}} - a_2 x_2^{\beta}$$

In this example vector field, $f_1(x) = x_2$ and $f_2(x) = -a_1 x_1^{\frac{\beta}{2-\beta}} - a_2 x_2^{\beta}$.

Let us consider a general dilation Δ_λ given as $(x_1, x_2) \mapsto (\lambda^{r_1} x_1, \lambda^{r_2} x_2)$. See that

$$f_1(\Delta_\lambda x) = \lambda^{r_2} f_1(x) \tag{5}$$
$$f_2(\Delta_\lambda x) = -a_1 \lambda^{\frac{\beta}{2-\beta} r_1} x_1^{\frac{\beta}{2-\beta}} - a_2 \lambda^{\beta r_2} x_2^{\beta} \tag{6}$$

For the equations (5) to be homogeneous, it is necessary that $\exists m$ s.t.

$$r_2 = m + r_1$$
$$\beta r_1/(2-\beta) = \beta r_2 = m + r_2$$

Solve these equations for r_1, r_2 and m. There are many solutions.

Ans: $m = (\beta - 1) r_2$, $r_1 = (2 - \beta) r_2$ where, r_2 is free parameter.

Homogeneous inclusions

Differential Inclusions

- Solution of systems with SMC are characterized in Filippov's sense.
- Filippov's method constructs a differential inclusion from the given discontinuous vector field.

Thus, the homogeneity of the inclusions needs to be defined.

Let a piece-wise continuous vector field f be homogeneous with degree m w.r.t a dilation $(r_1, \cdots, r_n)$, then Filippov's set-valued map associated with f is also homogeneous with degree m.

Now, we can state the theorem about finite-time stability of systems with discontinuous(piece-wise continuous) right-hand sides.

Theorem

Let the right-hand side of differential equation be, piece-wise continuous, homogeneous of degree $m < 0$, w.r.t. a dilation $(r_1, \cdots, r_n)$. If the differential equation is globally asymptotically stable then it is also globally finite time stable.

This is a very important and valuable result which connects asymptotic stability with finite time stability.

Proof of the convergence for twisting controller

Consider the uncertainty free version of the twisting controller,

$$\ddot{\sigma} = -k_1 sgn(\sigma) - k_2 sgn(\dot{\sigma}) \tag{7}$$

Proposition

The equilibrium point, origin, of (7) is globally finite time stable, i.e., σ and $\dot{\sigma}$ reach zero in finite time if the condition $k_1 > k_2 > 0$ is satisfied.

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The Lyapunov based approach for proof

- Show the asymptotic stability.
- Show that the vector field and corresponding set-valued map are homogeneous with a negative degree.

Represent the equation (7) in state variable form by defining $x_1 = \sigma$ and $x_2 = \dot{\sigma}$,

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -k_1 sgn(x_1) - k_2 sgn(x_2), \; k_1 > k_2 > 0$$

Lyapunov based proof (Orlov,2005)

The modern approaches to the finite-time convergence analysis of high-order sliding mode systems are based on the homogeneity principle.

Consider the Lyapunov function candidate as

$$V = k_1|x_1| + \frac{1}{2}x_2^2$$

Let us see if the function qualifies as a Lyapunov function.

- Positive definite? i.e., $V(x) > 0, \forall x \neq 0$ and $V(0) = 0$. Yes
- Continuously differentiable? No. It is not differentiable at $x_1 = 0$
- Negative definite/semi-definite time derivative along trajectories of (7).

The identity $\frac{d}{dy}|y| = sgn(y),\ y \neq 0$, has been used to find the derivative as follows

$$\dot{V} = k_1 x_2 sgn(x_1) - k_1 x_2 sgn(x_1) - k_2 x_2 sgn(x_2) = -k_2|x_2| \leq 0$$

The time derivative of candidate Lyapunov function is not strictly decreasing. There exists a set $\Omega = \{x \in \mathbb{R}^2 | x_2 = 0\}$, where the Lyapunov function is constant in time, i.e., the time derivative is zero.

LaSalle's invariance principle

The LaSalle's invariance principle is used to infer asymptotic stability in cases where the time derivative of Lyapunov function along system trajectory is not negative definite.

Let us see LaSalle's principle for the continuously differentiable vector field,i.e., $\dot{x} = f(x)$, with $f \in \mathcal{C}^1$

Let $\Omega \subset \mathbb{R}^n$ be a positively invariant, compact set. Let $V : \mathbb{R}^n \to \mathbb{R}$, be a continuously differentiable function such that $\dot{V}(x) \leq 0$ in Ω. Let E be the set of all points in Ω, such that $\dot{V}(x) = 0$. Let M be the largest invariant set in E. Then every solution starting in Ω approaches M as $t \to \infty$.

Asymptotic stability: The largest invariant set contained in E is only equilibrium point.

Discontinuous systems with non-smooth Lyapunov function

This principle cannot be used directly in our problem because of the following reasons:

- Our vector field is discontinuous.
- V is not continuously differentiable.

Extension of LaSalle's principle

Extension

- There exists a Lipschitz continuous, positive definite Lyapunov function V(x) such that its time derivative along the system trajectory is negative semi-definite. Mathematically,
$$\dot{V}(x) \leq 0$$
- Let M be the largest positively invariant set contained within the set where $\dot{V}(x) = 0$.

Then, all trajectories of the system converge to M as $t \to \infty$.

Recalling the Lyapunov function and its derivative $\dot{V}(x) \leq -k_2|x_2|$.

- Here $E = \{x \in \mathbb{R}^2 | x_2 = 0\}$, which is the x_1-axis.
- At any point $(c, 0)$ in the set E, the velocities are

$$\begin{aligned}
\dot{x}_1 &= 0 \\
\dot{x}_2 &\in -k_1 sgn(x_1) + [-k_2 \quad k_2] \\
&\in \pm k_1 + [-k_2 \quad k_2] \\
&\in [-k_1 - k_2 \quad -k_1 + k_2] \text{ OR } [k_1 - k_2 \quad k_1 + k_2] = F(x)
\end{aligned}$$

Finite time stability

$$\dot{x}_2 \in [-k_1 - k_2 \quad -k_1 + k_2] \text{ OR } [k_1 - k_2 \quad k_1 + k_2] = F(x)$$

- Now observe that if $k_1 < k_2$, then $0 \in F$, thus providing $x_2(t) \equiv 0$ as a Fillipov trajectory with E.
- However, with $k_1 > k_2$, no point, on x_1-axis, except origin, is positively invariant because the $F(x)$ does not contain zero.
- Thus, the origin of the considered system is asymptotically stable with $k_1 > k_2 > 0$.
- After showing an asymptotically stable origin, we should check if the system is homogeneous with some dilation.

Finite time stability

Homogeneity

Consider the dilation $\Lambda : (x_1, x_2) \mapsto (\lambda^2 x_1, \lambda x_2)$. Recall the original vector field

$$f_1(x) = x_2$$
$$f_2(x) = -k_1 sgn(x_1) - k_2 sgn(x_2)$$

The vector field with the above dilation becomes, as shown below

$$f_1(\Lambda x) = \lambda x_2 = \lambda^{2-1} f_1(x)$$
$$f_2(\Lambda x) = \lambda^{1-1} f_2(x)$$

Thus, the vector field is homogeneous with degree -1.

This proves that the origin of the twisting algorithm without uncertainties is finite time stable.

Thank You!